

User creates input file: 'config file'

These are the following parameters that are specified in the config file:

Placement of initial bacteria

Bacteria motility force

Bacteria reproduction rate

Initial nutrient concentration

EPS production rate

Nutrient threshold for EPS production

EPS size

EPS-cell attraction range

Coefficients of friction between particles

Simulation boundary size

Maximum agent limit

Intermediate output interval. The user can specify for the program to add to the output file at set time intervals. Shorter intervals will increase the framerate during visualisation

User receives output file

Output files contains a copy of the input parameters, plus these details, for each time point:

- 1: All bacteria and EPS positions, sizes, and cluster number (which cluster they belong to)
- 2: Centre positions, and sizes of all bacteria and EPS clusters
- 3: Position of the centre of the colony
- 4: Diameter of the colony

User puts output file(s) into graphing module as input and selects output type:

Types 1 through 6 as described in the output

User also chooses which input parameters to display on the graph (Other than the x and y axes' values)

User receives graphs as image files from graphing module

Graphs can be a few things:

- 1: (Fig 3 & 6) Largest n EPS or bacteria clusters, in descending order by size. Can show multiple output files on one graph (and therefore compare based on inputs)
- 2: (Fig 4) Phase separation diagram: x-values are motility, y is depletion effects, and then all points are individual input files. Colours indicate average cluster size: blue: large orange: medium green: small Can be based on bacteria or EPS clusters
- 3: (Fig 7) Size of largest cluster (EPS or bacteria) at a specific time, and size of largest cluster containing those particles over time, for multiple input files
- 4: (Fig 8 & 11) Colony roughness over time for multiple input files.
- 5: (Fig 10) Distribution of distance of particles to the centre of the colony for each EPS and bacteria over time
- 6: (Fig 11) Sparseness of colony over time. "Sparseness is defined as $\frac{1}{A \cdot particles(Acolony)}$ where those are the areas taken up by particles and the total colony

